

1 Subsidizing the tragedy of the commons: Fuel subsidies
2 modify fishing behavior and drive overfishing

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Abstract

Fuel subsidies to fishers are widely considered one of the leading drivers of over-fishing, prompting global calls for reform. Yet it remains unclear how fishing effort and landings would change in the absence of these subsidies. We address this question by analyzing 14 years of high-resolution data on fisher-level fuel subsidy allocations, fishing activity, and production in Mexico’s shrimp trawl fleet. Leveraging year-to-year variation in Mexico’s subsidy policy, we find that receiving a fuel subsidy increases fishing effort by 40.6%, with similar responses for fished area and landed catch. Mexico’s abrupt subsidy elimination in 2020 led to a 30% reduction in fleet size and decreased fishing time among remaining vessels. Together, these results show that fuel subsidies meaningfully increase fishing pressure and support ongoing global efforts to eliminate fuel subsidies.

Significance: International agreements—including SDG Target 14.6 and Target 18 of the Kunming–Montreal Global Biodiversity Framework—call for major reforms to fisheries subsidies. After decades of negotiation, the WTO has reached a global deal to curb harmful subsidies, yet the behavioral consequences of such reforms remain uncertain. Our study fills this gap by providing causal estimates of how fishers respond to fuel subsidies. The result inform debates about the likely effects of global reform and, together with related work, suggest that eliminating capacity-enhancing subsidies would substantially reduce fishing effort—both by inducing the exit of marginal vessels and by lowering effort among those that remain active..

1 Introduction

Fisheries subsidies are thought to be one of the leading causes of fisheries over-exploitation [1]. Scientists, practitioners, and politicians worldwide have called for eliminating or reducing fisheries subsidies as part of a global concerted effort to rebuild fish stocks [2]. However, our ability to predict the social and environmental outcomes of such reform hinges on the answer

to two crucial and as-of-now unanswered questions: How much additional fishing effort is caused by fuel subsidies? And, where does this additional effort, if any, manifest in the world? If the amount of overfishing induced by fuel subsidies is large, then the reforms could have large upsides. However, if the amount is small relative to other sources of overfishing (*e.g.* by-catch or illegal, unreported, and unregulated fishing), then it may be better to focus management efforts on addressing those. Similarly, if most “subsidized” effort is spatially located in some regions or fishing grounds, then we would expect that only those fishing grounds would benefit from a reform [3]. Here, we use high-resolution vessel tracking data from Mexican shrimp trawlers and long-term administrative data on fisher-level subsidy allocations to show how fuel subsidies influence fishing intensity, location, and production in Mexico’s largest and most valuable fleet.

Subsidizing an input such as fuel generally leads to a socially inefficient over-use of that input. When the input usage creates an externality (like carbon emissions or overfishing [4, 1]), the subsidy leads to two sources of lost economic efficiency. The first is the usual cost associated with a market distortion, which arises because resources are being misallocated. The second is associated with greater production of the externality itself. This is an understudied topic, but is of pivotal importance to the sustainability of agriculture, fisheries, mining, and other natural resource use settings, and implicitly underpins ongoing policy efforts to curb subsidies in these sectors [2]. This paper focuses on this second type of deadweight loss in the context of fuel subsidies in industrial fisheries.

Fisheries subsidies are prevalent in most coastal nations, where governments typically spend enormous amounts on subsidy programs [1]. In 2018 alone, nations provided a total of USD \$35.4 billion in fisheries subsidies, USD \$7.7 billion of which were granted as fuel subsidies. These large figures have prompted calls for global subsidy reforms [2, 5], and a particular focus has been placed on ending cost-reducing and capacity-enhancing subsidies such as fuel subsidies. Although there is broad consensus about the potential threats and damages posed by fuel subsidies in fisheries, empirical evidence on their social and environ-

64 mental costs remains limited.

65 There are two empirical challenges to studying the effect of fuel subsidies on fishing
66 outcomes. The first one is data availability. One must observe subsidy allocations, fishing
67 behavior, and/or fisheries production through time, none of which are readily available. And
68 even when they are, a second challenge arises. Nations typically implement blanket subsidy
69 policies for all fishers in a region, which eliminates the quasi-experimental variation required
70 to establish causality in observational studies [6].

71 However, novel datasets and clever use of causal identification strategies have allowed
72 some to make commendable progress on this issue. For example, Wang et al. [7] studied
73 the interaction between fuel subsidies and vessel buyback programs in China. By observing
74 vessel-level subsidy allocations, they found that reductions in fuel subsidies accelerated the
75 rate at which fishermen retired their vessels. However, they only studied the extensive
76 margin effects of subsidies as they did not employ vessel-tracking data to measure intensive
77 margin effects. Conversely, recent work by Englander et al. [6] leveraged vessel-tracking
78 data to show that fuel subsidies to China’s distant water fishing fleet have a large impact on
79 the fleet’s fishing effort, and that biological overfishing could be greatly reduced in several
80 high seas regions if China were to reduce fuel subsidies. While they were able to study
81 the intensive and extensive margin effects of fuel subsidies, they did not directly observe
82 vessel-level subsidy allocations. Neither work studied the effects on fisheries production.

83 To the best of our knowledge, these studies by Wang et al. [7] and Englander et al.
84 [6] are the only ones to provide rigorous causal estimates of the effect of fuel subsidies on
85 fisheries. Important as their contributions are, they focus on China, a nation whose fishing
86 dynamics, magnitude, and scale are so different that most work on fisheries treats them
87 differently [8, 9]. This means their findings may not generalize to other fleets—particularly
88 those in developing nations—highlighting the need to study the effect of fuel subsidies on
89 fisheries in other regions. We contribute to this literature by focusing on Mexico, the world’s
90 13th largest fishing nation[9]. As we will explain in detail below, Mexico provides an ideal

empirical setting in which to study the effect of fuel subsidies on fishing behavior and fisheries production.

The paper proceeds as follows. section 2 provides key details on Mexico’s fuel subsidy program and explains our identification strategy. section 3 presents our results, which are discussed in section 4. A detailed account of our data and methods are provided in section 5.

2 Mexico’s Fuel Subsidy Program

Mexico’s well-developed fishing industry has a long-history of being subsidized by federal programs, particularly in the way of fuel subsidies [10, 11, 12, 13]. This section provides a brief overview of the main characteristics of Mexico’s fuel subsidy program.

Our period of study covers 2011 - 2024. During this time, Mexico’s fishery management agency (CONAPESCA) maintained a limited-entry roster of economic units (*i.e.* fishers or fishing companies) eligible to receive a fuel subsidy each year. An economic unit may only “enter” the roster if another unit “exits” it¹. Once in the roster, the amount of subsidy granted to an economic unit is a function of a vessel’s engine power and the prescribed duration of the fishing season. However, fishery managers can adjust the final subsidy amount each unit receives based a factor constant across all units that depends on the annual national allocations to the program (For more details, see Supplementary Materials). Subsidized economic units receive money via a government-issued debit card, which can only be used at fueling depots. Any unspent money at the end of the year is reclaimed by CONAPESCA.

The program was suddenly eliminated in 2020. In late 2019, Mexico’s former president signed into law the “Republican Austerity Law”, which aimed to curtail public spending. Almost at the same time, in early 2020, the World was faced with the COVID-19 pandemic. Together, these two factors forced Mexico to completely eliminate fuel subsidies without no-

¹For example, an economic unit could be removed from the roster if they failed to carry a functioning vessel monitoring system or if they failed to report their landings.

115 tice [13]. Although this led to multiple protests from fishers and increased political pressure²,
 116 the federal government did not budge and, since then, Mexican fishers have not received sub-
 117 sidized fuel.

118 We seek to understand how fuel subsidies modify the *magnitude* and *footprint* of fishing
 119 because these determine the upsides of a reform. As an example to motivate our analysis,
 120 Figure 1 shows how fishing activity by one economic unit changes when they receive a
 121 fuel subsidy of MXN \$231,543 (about USD \$12,388). When subsidized, the economic unit
 122 increases fishing hours (from 513 hrs/yr to 2,880 hrs/yr) and the extent of its fishing grounds
 123 (from 8,395 Km² to 12,572 Km²). This example from a single economic unit in our data
 124 cannot account for other factors that might explain the changes observed. However, it hints
 125 as to how fuel subsidies modify the intensity and location of fishing.

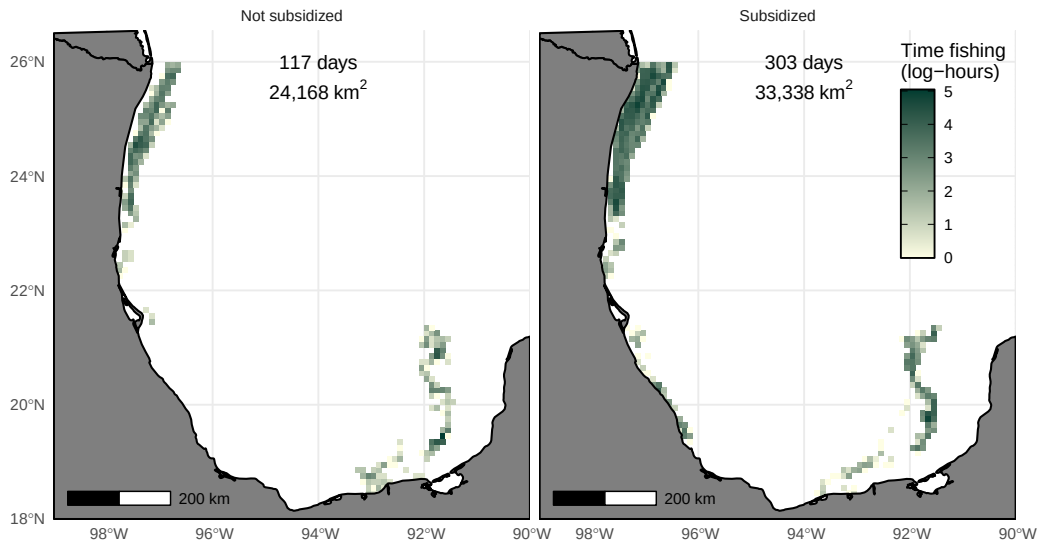


Figure 1: Example of changes fishing behavior in relation to subsidy status. Maps show fishing activity by the same economic unit in a year without a subsidy (left) and a year with a subsidy (right). This fisher spent nearly three times more time fishing when subsidized than when not, and the extent of its fishing grounds is around 50% larger when subsidized than when not. The footprint of fishing effort is shown along a 0.1° grid.

²See El Sudcaliforniano: Pega a pescadores la falta de apoyos for a news report and a letter by Senator Cecilia Sánchez García denouncing the removal of fisheries subsidies.

2.1 Identification Strategy

Our empirical strategy leverages three complementary sources of quasi-experimental variation in Mexico’s fuel subsidy program. Together, these provide plausibly exogenous shocks that allow us to estimate how fuel subsidies shape fishing behavior and fisheries production along both the extensive and intensive margins. Here, we summarize the core logic behind each source of variation and the corresponding empirical estimators.

Roster turnover and changes in subsidy status. First, we exploit changes in subsidy status generated by the institutional design of CONAPESCA’s subsidy roster. Entry into the roster can only occur when another economic unit exits. Because an economic unit cannot influence whether others exit the roster, these transitions effectively induce externally driven shifts between *subsidized* and *not subsidized* states for units that are subsidized only in some years. We use this roster turnover to identify how receiving a subsidy affects the probability of fishing, the probability of reporting landings, and the magnitude of fishing activity conditional on participation in the fishery. These effects are estimated using two-way fixed effects models that capture semi-elasticities with respect to subsidy status.

Within-unit variation in subsidy amounts. Second, we leverage within-economic-unit variation in subsidy amounts, which arises from annual adjustments to the program’s “adjustment factor.” This factor is determined by federal budget allocations and administrative decisions that vary across regions and years but are orthogonal to individual fishers’ short-run incentives or expected fishing behavior. As a result, the year-to-year fluctuations in subsidy amounts for the same economic unit provide treatment-intensity variation that is plausibly exogenous. For example, Figure 2 shows how economic-unit-level subsidy allocations changed between 2011 and 2019, when Mexico provided fuel subsidies to its fleet. We exploit this within-unit, across-year variation using log-log panel regressions with economic-unit and region-year fixed effects, yielding elasticities of fishing time, fished area, and landed

151 catch with respect to the amount of subsidy received.

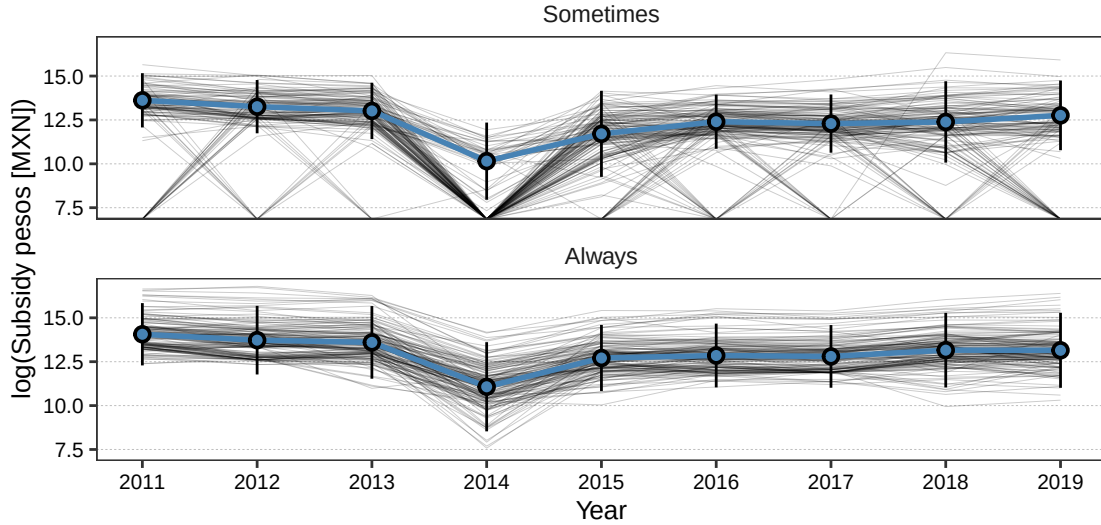


Figure 2: Change in the subsidy amount (Mexican Pesos) granted to each economic unit between 2011 and 2019. The top panel shows data for economic units that are subsidized at least once, the bottom panel shows data for economic units that are always subsidized in our period of study. Each thin black line corresponds to one economic unit. When a line touches the horizontal axis it implies it is not subsidized in that period. The overlaid points show mean $\pm$ sd. Note that not a single black line is perfectly horizontal, providing evidence of within-unit variations in subsidy allocations through time.

152 **Abrupt elimination of subsidies in 2020.** Third, we exploit the sudden and unan-
153 ticipated elimination of fuel subsidies in 2020 as a natural experiment. The combination
154 of the 2019 Republican Austerity Law and fiscal pressures associated with the COVID-19
155 pandemic resulted in the complete removal of subsidies without advance notice. Because the
156 reform was nationwide, non-discretionary, and unrelated to the performance or behavior of
157 individual economic units, it provides a clean policy shock. We use this setting to estimate
158 (i) the probability that an economic unit exits the fishery after subsidies disappear and (ii)
159 changes in fishing behavior among units that remain active. These effects are estimated with
160 event-study models and two-way fixed effects regressions.

161 Together, these three sources of variation allow us to estimate: (i) semi-elasticities by
162 comparing subsidized and non-subsidized states; (ii) elasticities using continuous changes in
163 subsidy amounts; and (iii) long-run and aggregate responses using the 2020 reform. This

multi-layered identification strategy provides a comprehensive assessment of how fuel subsidies affect fishing behavior, spatial footprint, and fisheries production.

Mexico provides a data-rich environment conducive to this study. Using different sources of administrative data, we built an annual panel of subsidy allocations (MXN \$), time fishing (hours), area fished (km^2), and landed catch (kg) by economic unit. This panel allows us to study changes in fishing behavior and production of 920 shrimp trawling vessels owned by 341 economic units. We focus on Mexico’s shrimp trawl fleet because they are Mexico’s largest and most important fleet, accounting for more than 45% of the value of all Mexican fisheries and catching around 50,000 tons each year [14, 15, 16]. Fuel represents around 4% of the fleet’s variable costs, and subsidies make were the fleet’s main source of profit [17, 13]. The fleet also received between 48% and 67% of the annual fuel subsidies provided to all industrial economic units fishing in Mexico.

A visual inspection of our panel data set reveals three general patterns (Figure 3). First, economic units spend more time fishing, fish a greater area, and land more shrimp than vessels that are not subsidized. Second, economic units that were always subsidized between 2011 and 2019 ($N = 142$) consistently fish more than those that were only sometimes subsidized ($N = 167$), and *vice versa*. And third, that the pattern persists even for the subset of economic units whose subsidy status changes in time within our sample (labeled “sometimes”).

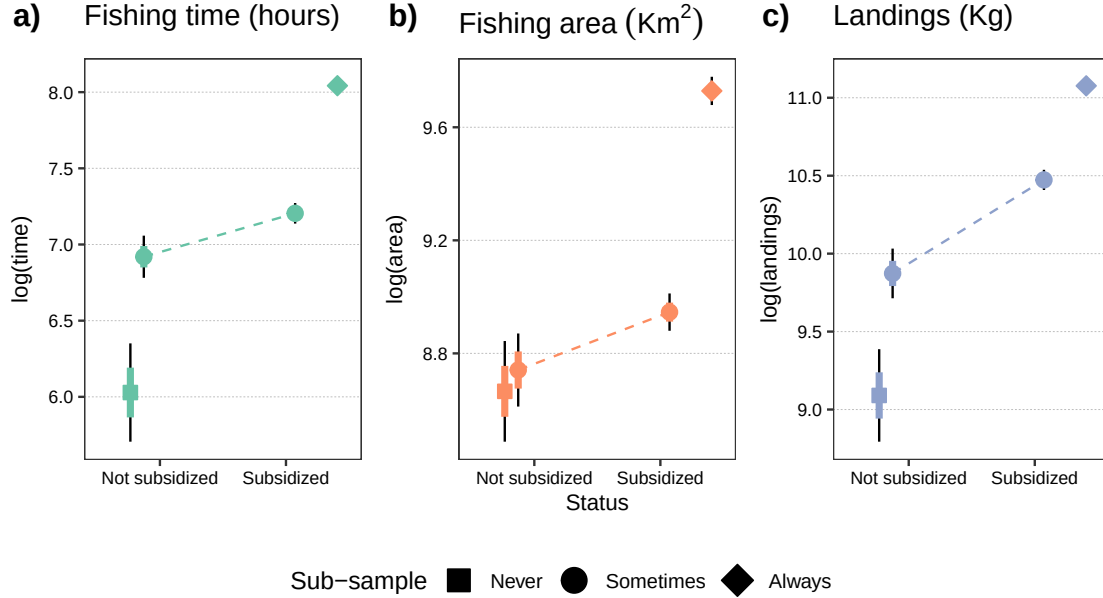


Figure 3: Fishing behavior and fisheries production in relation to subsidy status (2011-2019). The horizontal axis shows the subsidy status and the vertical axis shows the log-transformed outcome of interest (time fishing, area fished, and landings). Points show mean values, error bars show standard errors (colored portion) and 95% confidence intervals (thin black lines). Marker shapes indicate subsidy category with respect to number of times subsidized (never, sometimes, always). The dashed lines connect the mean values for economic units that are sometimes subsidized across subsidized status.

3 Results

We structure our results around the three sources of variation described in subsection 2.1. First, we examine how changes in subsidy status affect fishing behavior and production. We then quantify how these outcomes respond to the amount of subsidy received. Next, we study the effects of Mexico’s 2020 reform. Finally, we combine these estimates to measure the aggregate and spatial impacts of fuel subsidies.

3.1 Responses to change in subsidy status

Our sample contains 341 economic units targeting shrimp between 2011 and 2024. Of these, 32 units never received a subsidy, 142 were always subsidized, and 167 were subsidized only in some years. We exploit within-unit changes in subsidy status for this last group to estimate how receiving a subsidy affects participation, fishing activity, and landings.

194 We begin by examining the extensive margin. Using a two-way fixed-effects framework
195 (see Methods), we find that receiving a subsidy increases the probability that an economic
196 unit engages in fishing by approximately 20%, as measured by both fishing activity and
197 area fished. The probability of reporting landings increases by 41% ($p < 0.01$; Table 1A).
198 These results indicate that some economic units participate in the fishery—and report their
199 catch—only when subsidized.

200 We then examine intensive-margin responses. On average, subsidized units fish 350
201 additional hours per year ($p < 0.01$), expand their fishing grounds by 2,528 km² ($p < 0.01$),
202 and land 3.2 additional tons of shrimp ($p < 0.01$). Relative to the mean outcomes of
203 unsubsidized units, these correspond to increases of 24.8%, 36.5%, and 170% in time fishing,
204 fishing area, and landings, respectively (Table 1B).

205 Finally, we estimate semi-elasticities for economic units that fish regardless of their
206 subsidy status. Receiving a subsidy is associated with a 40.69% increase in time fishing
207 ($p < 0.01$), a 23.5% expansion in fishing grounds ($p < 0.01$), and a 70.18% increase in land-
208 ings ($p < 0.01$; Table 1C). All results are robust to alternative specifications and sample
209 definitions (see Table S2–Table S4 and Figure S3–Figure S8).

Table 1: Effect of receiving a fuel subsidy on fishing behavior and fisheries production.

	Fishing time	Fishing area	Landings
A) Extensive margin			
Subsidized	0.223 (0.028)***	0.219 (0.029)***	0.413 (0.037)***
N_{eu}	167	167	167
N	1431	1431	1431
R^2 Adj	0.531	0.547	0.597
B) Intensive margin (levels)			
Subsidized	350.248 (65.644)***	2528.599 (523.001)***	32347.423 (6977.661)***
$\bar{Y}_{\text{Subsidized}=0}$	1411	6939	18968
N_{eu}	167	167	167
N	1431	1431	1431
R^2 Adj	0.924	0.792	0.710
C) Semi-elasticities			
Subsidized	0.341 (0.073)***	0.211 (0.068)***	0.532 (0.080)***
N_{eu}	134	127	117
%Change	40.69%	23.50%	70.18%
N	1290	1235	1192
R^2 Adj	0.726	0.700	0.757

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows estimates for the extensive margin, where the outcome variables indicate whether a vessel spent time fishing, had fishing grounds, or reported landings. Panel B) shows estimates for the intensive margin, where the outcome variables are time fishing (hours), fishing area (km²), and landings (kg). Panel C) shows semi-elasticity estimates for log-transformed time fishing (hours), fishing area (km²), and landings (kg). This last panel excludes observations with fishing activity or landings at exactly zero, mostly capturing the intensive margin.

3.2 Responses to change in subsidy amount

The effects above capture differences between subsidized and unsubsidized states. We now examine how fishing effort, spatial footprint, and landings respond to differences in the *amount* of subsidy received.

A key feature of Mexico’s fuel subsidy program is that the annual subsidy amount allocated to each economic unit varies across years (see Figure 2). This variation arises from changes in federal budget allocations and administrative decisions within CONAPESCA, including allocations to other programs such as aquaculture subsidies [11, 13]. Because these adjustments are driven by budgetary conditions rather than unit-specific factors, they generate plausibly exogenous within-economic-unit variation in subsidy amounts. We use this year-to-year variation to estimate intensive-margin responses for the 297 economic units that were subsidized at least twice between 2011 and 2019.

Using a two-way fixed-effects regression, we estimate elasticities of fishing time, fished area, and landings with respect to subsidy amount. We find that a 1% increase in the subsidy received is associated with a 0.14% increase in time fishing ($p < 0.01$), a 0.15% increase in fished area ($p < 0.01$), and a 0.20% increase in landings ($p < 0.01$). These estimates are robust to alternative sample definitions. Specifications that substitute covariates for fixed effects produce 40–80% larger elasticity estimates, but the qualitative patterns remain unchanged (see Table S5, Figure S9, and Figure S10).

Table 2: Elasticity estimates for time fishing (hours), fishing area (km²), and landings (kg) with respect to changes in subsidy amount.

	Fishing time	Fishing area	Landings
log(subsidy amount[MXP])	0.139 (0.021)***	0.147 (0.022)***	0.198 (0.018)***
%Change	0.14%	0.15%	0.20%
N_{eu}	297	292	295
N	2240	2202	2246
R^2 Adj	0.850	0.818	0.876

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. All models include fixed effects by economic unit and by region-year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. The sample contains economic units subsidized at least twice and with subsidy amount > 0 . The number of economic units used in each column is shown by N_{eu} .

3.3 Responses to an *impromptu* reform

While the previous subsections rely on within-program variation, Mexico’s abrupt 2020 reform provides an opportunity to study how economic units respond when subsidies disappear entirely. The COVID-19 pandemic interacted with an ongoing austerity plan, resulting in sweeping fiscal and administrative adjustments [12, 13]. The fuel subsidy program was eliminated practically overnight, prompting protests from fishers who argued they could not continue operating without support [18].

We begin by examining exit behavior. We focus on the 142 economic units that were always subsidized prior to the reform. All were active between 2011 and 2019, yet 53 failed to participate in the fishery at least once between 2020 and 2024, and 42 did not return at all by 2024, suggesting permanent exit.

Estimating the probability of exit after 2020, we find that the average annual probability of an economic unit leaving the fishery increased by 7.5 percentage points in the post-reform period ($p < 0.01$; Table 3). By 2024, the cumulative exit probability reached approximately 30% (Figure 4).

We next examine how remaining economic units adjusted their behavior. Among units that continued fishing after the reform, we estimate modest reductions along the extensive

Table 3: Effect of Mexico's *impromptu* fuel subsidy reform on probability of economic units exiting the fishery.

	(1)
Post	0.075 (0.011)***
N	1914
R^2 Adj	0.049

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level.

margin (Panel A of Table 4). Intensive-margin responses were substantially larger: average annual fishing time fell by 4,200 hours, fished area declined by 13,900 km², and landings decreased by 45.6 tons (Panel B of Table 4). Event-study estimates (Figure 4) show that these changes stabilized around 2022.

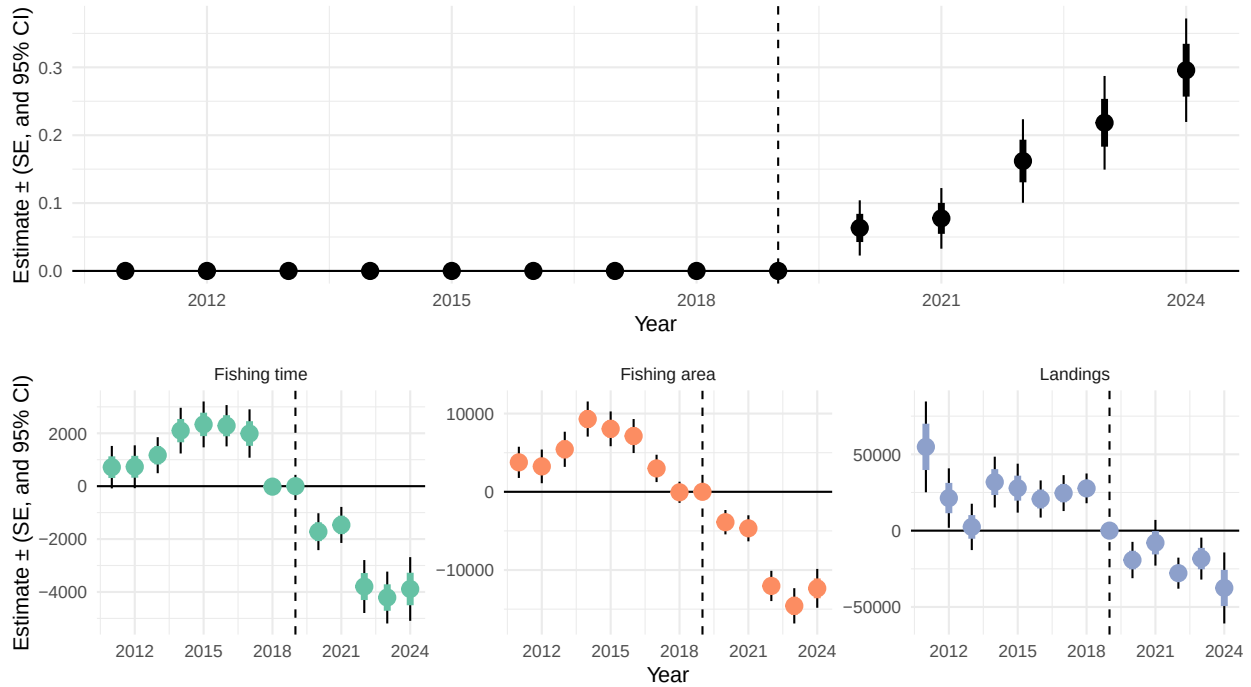


Figure 4: Annual estimates of the effect of Mexico's fuel subsidy reform on fisheries. The top panel shows changes in the probability of exiting the fishery for economic units that were always subsidized between 2011 and 2024. The bottom panels show changes in total fishing time, fishing area, and landings for economic units that did not exit the fishery as of 2024. Points are coefficient estimates, thick colored lines show standard errors, and thin black lines show 95% confidence intervals.

Table 4: Effect of Mexico’s *impromptu* fuel subsidy reform on fishing behavior and fisheries production for economic units that did not exit the fishery.

	Fishing time	Fishing area	Landings
A) Extensive Margin			
Post	-0.036 (0.012)***	-0.093 (0.021)***	-0.086 (0.021)***
N	1400	1400	1400
R^2 Adj	0.099	0.185	0.188
B) Intensive Margin (levels)			
Post	-4273.615 (685.285)***	-13931.084 (1112.424)***	-45651.016 (10855.586)***
$\bar{Y}_{\text{Post}=0}$	4472	18351	107863
N	1400	1400	1400
R^2 Adj	0.783	0.748	0.868

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows estimates for the extensive margin, where the outcome variables indicate whether a vessel spent time fishing, had fishing grounds, or reported landings. Panel B) shows estimates for the intensive margin, where the outcome variables are time fishing (hours), fishing area (km²), and landings (kg).

3.4 Aggregate effects of fuel subsidies

We have shown that subsidized economic units fish more, and that the amount of additional fishing increases with the amount of subsidy received. We now examine what these *individual* responses imply for *aggregate*, fishery-wide impacts. In particular: How much of total historical effort is attributable to fuel subsidies? Because vessels are heterogeneous in behavior and spatial footprint, and subsidies are not evenly distributed, we combine yearly vessel-level data on subsidy receipt, subsidy amount, and fishing location to quantify the share of historical fishing activity and landings attributable to subsidies. We then identify areas disproportionately subject to subsidized effort.

3.4.1 Historical impacts of subsidies

Total annual fishing activity (or landings) can be divided into three categories: (1) activity by units that were never subsidized; (2) activity by subsidized units that would have occurred

even without the subsidy; and (3) activity by subsidized units that is attributable to the subsidy. Between 2011 and 2019, Mexican shrimp trawlers fished an average of 1.13 ± 0.15 million hours, of which 1.05 ± 0.15 million were by subsidized units (Figure 5a). Using our semi-elasticity estimates, we find that 0.42 ± 0.06 million hours per year can be attributed to fuel subsidies, implying that subsidies accounted for $37.6\% \pm 2.4\%$ of total annual fishing hours.

The fleet landed an average of 17.6 ± 1.6 thousand tons of shrimp per year, with 16.9 ± 1.8 thousand tons landed by subsidized units (Figure 5c). Our estimates indicate that 11.9 ± 1.2 thousand tons of landings—about 67% of all shrimp landed by the studied fleet—were attributable to subsidies.

We repeat this exercise using our elasticity estimates to calculate the *percent* reduction in fishing time and landings under alternative subsidy-reduction scenarios (10, 30, 50, and 90%). For instance, capping subsidy allocations at 50% of historical levels would have reduced fishing time by approximately 96.3 thousand hours per year and landings by 2.18 thousand tons (Figure 5b,d).

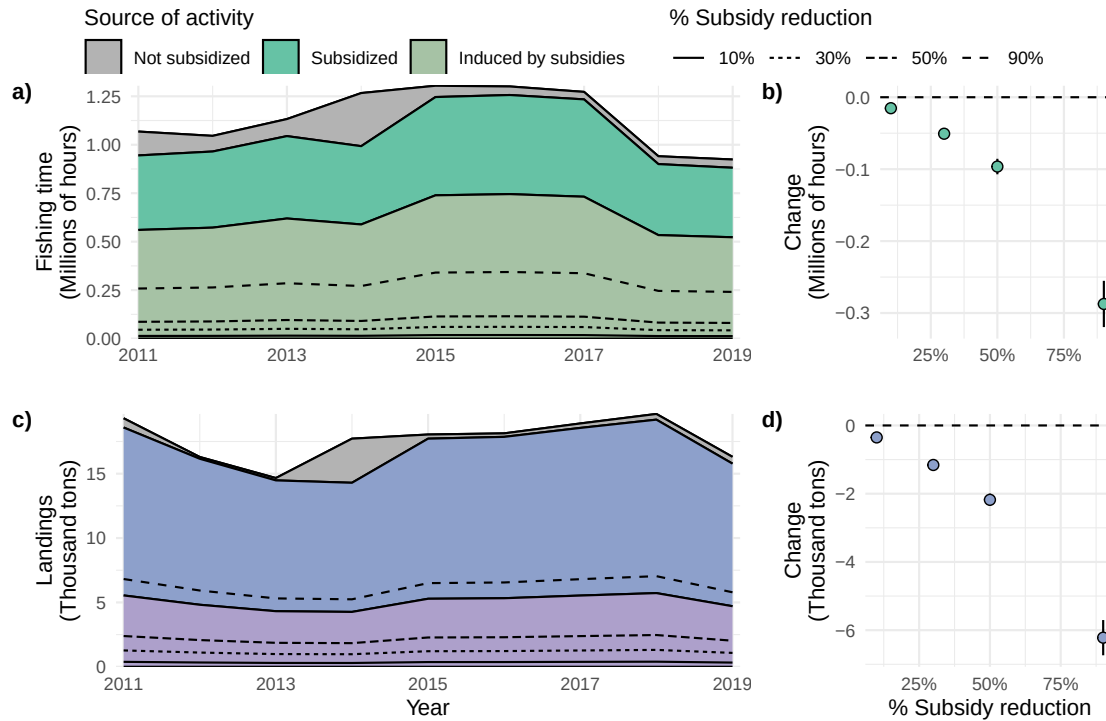


Figure 5: Aggregate effects of fuel subsidies on fishing activity, fished area, and landings. Panels a) and c) show area-stacked time-series of fishing time and landings³. The gray portion is activity and production from economic units that were not subsidized in a given year. The colored portion corresponds to activity and production by economic units that were subsidized. The bottom stack of each panel shows the portion of effort or production by subsidized economic units that is attributable to the subsidy, as indicated by our semi-elasticity estimates (Table 1C). The different line types show the portion of effort that could have been removed had the subsidies been reduced by different amounts. Panels b) and d) show the mean annual reduction in fishing time and landings expected from four different subsidy reduction policies (estimated as mean of all activity between 2011-2019). Black error bars show 95% confidence intervals and the colored portion shows standard errors.

3.4.2 Spatial implications of a non-spatial policy

The above thought experiments cannot be conducted for fished area because the metric is not additive across economic units. However, understanding the spatial implications of a non-spatial policy is crucial, as the distribution of fishing effort can strongly influence environmental impacts [19]. For example, if most economic units fish in a small subset of grounds, eliminating fuel subsidies would generate large, localized upsides. Conversely, if subsidized and unsubsidized vessels operate in broadly similar areas, subsidy reform would yield more modest but spatially widespread effects. This tension between large and local *versus* modest and widespread benefits raises a natural question: is subsidy-induced fishing

effort homogeneously distributed in space?

To answer this, we use our semi-elasticity estimates to simulate counterfactual fishing activity in the absence of subsidies along a $0.1^\circ \times 0.1^\circ$ grid (roughly 11 km by 11 km at the equator). Pixels fished only by unsubsidized units show no change, while those fished exclusively by subsidized units exhibit the largest reductions. Using data from the final year of subsidies (2019, with 286 active units, 247 of which were subsidized), we find that subsidized fishing activity is heterogeneously distributed in space, but that this pattern broadly matches the baseline distribution of activity by unsubsidized units (Figure 6a–b).

Mexico divides its waters into six broad management regions. The Gulf of California (region II) and Gulf of Mexico (region V) sustain the highest levels of both total and subsidized activity (Figure 6a–b). Subsidized effort accounted for up to 40% of total activity along the eastern margin of the Gulf of California and the Campeche Bank, between regions V and VI (Figure 6c). These areas also consistently ranked among the top 90% of pixels where subsidized effort represented a large share of total fishing (Figure 6d).

Finally, the analysis reveals that several areas of particular conservation concern were fished primarily by unsubsidized units. For example, all activity within and around the recently protected Alacranes Reef [20] (region VI) was attributable to unsubsidized vessels. Similarly, fishing in the upper Gulf of California—region II, habitat of the critically endangered Vaquita [21, 22]—was also predominantly conducted by unsubsidized units.

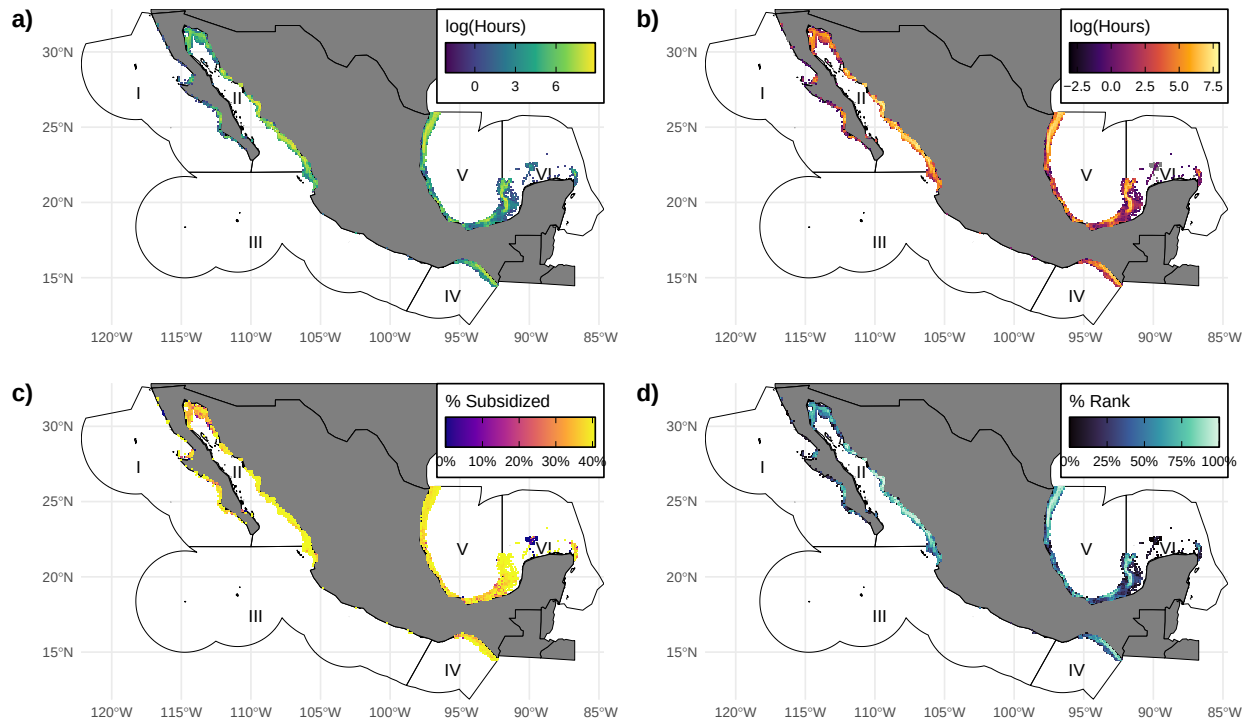


Figure 6: Spatial distribution of the effects of fuel subsidies on fishing activity. Panel a) shows a map of total fishing hours for 2019. Panel b) shows a map of total fishing hours attributable to fuel subsidies. Panel c) shows the per cent of fishing effort that is attributable to fuel subsidies, and panel d) shows the percentile ranking of each pixel. Polygons in the ocean show Mexico's Exclusive Economic Zones, divided into six management regions utilized by Mexico's fishery management agency.

4 Discussion

Our results provide clear evidence that fuel subsidies significantly increase fishing effort, spatial footprint, and landed catch in Mexico’s industrial shrimp fleet. First, receiving a subsidy induces large increases in time spent fishing, area fished, and landings. Second, the amount of subsidy matters: fishing effort, fishing grounds, and production all scale positively with the size of the subsidy allocation. Third, Mexico’s abrupt 2020 reform led to substantial exit from the fishery and to marked reductions in effort and landings among vessels that remained active. Taken together, these findings indicate that fuel subsidies have meaningful and measurable fishery-wide impacts.

These results align with long-standing theoretical expectations that subsidizing variable inputs leads to higher effort and greater extraction, but empirical confirmation has been limited. Our study provides such confirmation using high-resolution vessel-tracking data, fisher-level subsidy allocations, and a set of quasi-experimental natural experiments unique to Mexico’s fishery management system. The Agreement on Fisheries Subsidies at the World Trade Organization, which entered into force in 2025, aims in part to curtail precisely these capacity-enhancing subsidies. Our results suggest that reforming fuel subsidies can meaningfully reduce overfishing pressure in industrial fleets.

No observational study is without limitations. We interpret our estimates as causal effects of subsidy allocation on short-run fishing behavior for two reasons. First, annual subsidy amounts are largely determined by federal budget allocations rather than fleet-level economic incentives. Second, fishers have limited ability to manipulate the observation of their behavior, as VMS transponders operate independently of reporting incentives. However, our estimates for landings should be interpreted with some caution, as landings reflect both biological catch and reporting behavior, the latter of which is tied to eligibility for future subsidies.

Our results show that subsidizing fuel alters fishing activity, but how managers allocate and disburse subsidies also shapes the magnitude of fishers’ responses. Mexico’s program

limits the quantity of subsidized fuel any fisher can obtain because, although there is considerable year-to-year and fisher-to-fisher variation, the allocation rule establishes a 2-peso per liter price subsidy over the first 40–70% of anticipated fuel consumption (DOF 2010–2018). These subsidies are disbursed as lump-sum transfers that can only be used for fuel. Fishers use this cash to pay for fuel until funds are exhausted (*i.e.*, the first consumed liters are effectively “free”). This price structure resembles an increasing block rate pricing scheme, often used for electricity and water. In those markets, consumers respond to the “average price” rather than the marginal price [23]. Using the allocation rule and a median diesel price of 16.2 pesos per liter, we calculate that Mexico’s fuel subsidies reduce the average price of fuel by 4.9–8.6%, consistent with other estimates (*e.g.*, 8.2% in Revollo-Fernández et al. 12). Our empirical results suggest that this is sufficient to induce a behavioral response.

Our aggregate calculations show that up to 37% of historical fishing effort is attributable to subsidies, with some areas (*e.g.*, the Campeche Bank and eastern Gulf of California) disproportionately affected. These observations imply that subsidy reform might generate large but localized environmental benefits. Limited availability of stock assessment data precludes precise statements about all relevant stocks, but we can put these numbers in context. For example, the biomass of the heavily fished blue shrimp (*Litopenaeus stylirostris*) stock in the Gulf of California [24] is estimated at 30% below the biomass that would yield MSY (*i.e.*, $B/B_{MSY} = 0.7$). It is therefore reasonable to expect that reducing fuel subsidies would support stock rebuilding, at least along the eastern Gulf of California.

We also show that areas of particular conservation concern (such as Alacranes Reef and the Upper Gulf of California) are primarily targeted by unsubsidized units. This suggests that subsidy reform alone would have limited direct implications for these areas. Other management measures, such as fully protected marine reserves, may be more appropriate when the objective is to reduce fishing pressure over sensitive habitat.

Overall, our findings suggest that subsidy reforms elsewhere may produce spatially uneven responses, with some areas benefiting more than others. It is also important to consider

social implications, as some ports or fishing communities may depend more heavily on subsidies. Previous work in Mexico and elsewhere has shown that even ideal management aimed at maximizing long-term yields may not raise fishers' income above the poverty line [25, 26]. Some have suggested reallocating funds from harmful fuel subsidies toward social programs intended to raise fishers' income [27], although practical pathways remain unclear. This tension between biological gains and political costs may help explain many nations' hesitation to reform fisheries subsidies and highlights the importance of understanding the distributional implications of reform.

We conclude that fuel subsidies induce overfishing, that the magnitude of overfishing is non-trivial, and that its spatial effects are highly localized. These findings support calls for subsidy reform [2] but also underscore the need for realistic expectations. Our results are directly relevant to Mexico and to other coastal nations considering reducing or eliminating fuel subsidies to their industrial fishing fleets.

5 Methods

We are interested in studying how fuel subsidies affect fishing activity and production. Here, our unit of observation are “economic units”, a term used by the Mexican fisheries agency (CONAPESCA) to refer to an individual or firm that participates in a fishery. We combine administrative datasets on subsidy allocations by economic unit and vessel tracking data to construct a unique panel of annual subsidies and fishing activity by economic unit. The following subsections provide further details on data procurement, filters, and sample construction.

5.1 Datasets and their sources

We make use of six types of data to study the effects of fuel subsidies on fishing behavior. Historical subsidy allocations and a vessel registry allow us to identify subsidized economic units and their characteristics. Then, we use vessel tracking data and historical fisheries production data to derive our three main outcomes of interest: fishing time, fishing extent, and fisheries production. Finally, we also use historical diesel fuel prices and monthly indices for El Niño Southern Oscillation to include fuel costs and environmental variation as covariates. Each of these is described in detail below.

5.1.1 Subsidy allocations

Data on subsidy allocations to each economic unit come from CausaNatura, an NGO whose mission is to compile, procure, and make available administrative datasets relevant to environmental and natural resource management. We use the “*Padrón de beneficiarios de Combustibles*”, which was last updated on June, 2020. This administrative dataset contains information on the annual subsidy cap assigned to economic units fishing in Mexico during the 2011 - 2019 period ($n = 4,597$). From this information, we assign treatment status (subsidized or not subsidized) to all economic units in our sample (see subsection 5.2), and the

amount of fuel subsidy received by each.

5.1.2 Vessel registry

We use an official vessel registry with information for all large-scale fishing vessels that hold a fishing permit in Mexico, which was also provided by CausaNatura. The vessel registry includes unique vessel identifiers and economic unit identifiers (ownership), vessel dimensions (length overall, beam, draught, and gross tonnage), species-specific fishing permits granted, and engine characteristics (*e.g.* total engine power, type of fuel used by the engine, and engine model). The registry contains information for 3,093 vessels owned by 1,093 economic units. From these, 1,415 are licensed to use bottom trawl nets and 1,561 are licensed to participate in the shrimp fishery; 1,368 are licensed to use both (and are owned by 464 economic units).

5.1.3 Vessel tracking data

[28, 29] There are two general types of vessel tracking technologies: Automatic Identification Systems (AIS) and Vessel Monitoring Systems (VMS). AIS is designed as a vessel-to-vessel broadcast system intended to help avoid at-sea collisions between vessels [8]. VMS, on the other hand, is employed by governments to track vessels of interest, and a vessel’s position is broadcast directly to a central repository instead of to other vessels [30]. We use VMS data from Mexico’s satellite monitoring system of fishing vessels (*i.e.* SISMEP[28]). These data are publicly available and continuously updated at datos.gob.mx. The version we use was downloaded on June 15, 2024. These VMS data contain the timestamp, geographic location (latitude and longitude), and speed of 2,775 vessels between January 1, 2007 and Feb 29, 2024.

Regulations require all fishing vessels larger than 10.5 m in length overall and with an in-board motor of more than 80 horsepower to carry a Vessel Monitoring System (VMS)⁴.

⁴Regulatory text available at: <https://www.monitoreodeembarcaciones.com.mx/monitoreosatelital/QuienDebe.htm>

Failure to comply with this VMS requirement automatically disqualifies a vessel as eligible to receive any type of subsidy.

5.1.4 Fisheries production

[31, 32] Fisheries production data come from Mexico’s landing receipts, where fishers report their landings. As with the VMS requirement, failure to report catch makes a fisher ineligible to receive a subsidy. The dataset contains information on the identity of the economic unit and vessel landing the catch, the target species, and the amount (Kg).

5.1.5 Fuel prices

We also compile price data for diesel fuel used by these economic units by combining two sources of information. The first one is reported by the Energy Information System (“Sistema de Información Energética”; “SIE”) and contains the national annual average price of diesel between 2011 - 2016, when fuels were subject to nation-wide price controls. Price controls were lifted in 2017, and fuel prices were determined by local supply and demand. The Energy Regulatory Commission (“Comisión Reguladora de Energía”; CRE) reports monthly state-level prices after 2017, which we use to calculate annual national averages for 2017 - 2019 period. We use Mexico’s consumer price index reported by the OECD to adjust prices to 2019 Mexican pesos.

5.1.6 Environmental covariates

The productivity of shrimp fisheries is known to be influenced by ENSO events [33]. We use the Mean NINO 3.4 index available from NOAA’s Physical Sciences Laboratory Climate Indices repository (Monthly Atmospheric and Ocean Time Series). We use monthly means to produce an annual mean value of ENSO 3.4, which we include as a time-varying covariate in some of our regressions.

5.2 Data processing

5.2.1 Sample construction

We limit our data to activity occurring between 2011 and 2019, the years for which subsidy allocation data are available. Additionally, retain vessel tracks occurring at depths between 0.15 and 100 m deep (as indicated by GMEDs bathymetric dataset) because shrimp trawlers in Mexico are not allowed to fish shallower than 9.15 m deep and they operate at a maximum depth of 100 m [14]. Shrimp trawlers typically operate speeds between 1 and 5 knots⁵, so we also filter tracks based on their speed. These filters result in a total of 1,177 vessels belonging to 414 economic units. We further restrict the sample to economic units that are only licensed to fish for shrimp using trawl nets, leaving us with 409 economic units.

5.2.2 Outcomes of interest

Our first outcome of interest is fishing activity. We define it as time (hours) a vessel spent traveling at speeds between 1 and 5 knots in areas between 9.15 and 100 m depth. We calculate an economic unit's total annual fishing hours as the sum of fishing hours across all their vessels.

Our second outcome of interest is the total extent of fishing grounds (km²) in which these economic units operate. We used a density-based spatial clustering algorithm (DBSCAN) to identify fishing grounds based on individual vessel positions. The algorithm was applied to all positions at the vessel-by-year level. The algorithm clusters points based on their distribution across space, given a minimum number of points per cluster and a maximum distance between points. We used a maximum distance of 50Km and a minimum of 6 points per cluster. Clusters thus represent the group of individual GPS coordinates that are associated with a fishing ground. Points without cluster membership were dropped. We then built a convex hull around each cluster and calculated its area. The total extent of fishing

⁵Catálogo de los Sistemas de Captura de las Principales Pesquerías Comerciales, available at: CONAPESCA

grounds of an economic unit was then calculated as the sum of all fishing grounds used by their vessels. For this portion of the analysis, geographic coordinates were reprojected onto a Mexico Lambert Conic Conformal projection (With EPSG code 6361).

Our third and last outcome of interest is the total amount of catch landed by each economic unit, which we derive from our fisheries production dataset. Our sample is therefore made up of large-scale economic units that target shrimp and carry VMS transponders. This group receives between 48.22% and 67.73.% of the annual subsidies awarded to all industrial economic units fishing in Mexico. The final estimation sample is a panel of annual economic-unit fuel subsidy allocations (in 2019 MXP), time, extent, landings, and control variables such as fuel prices, total horsepower of number of active vessels owned by an economic unit, and environmental indices (*i.e.* NINO3.4 index). These data contain 3,376 observations attributed to 409 economic units between 2011 and 2019. Tables with summary statistics are included in the supplementary materials (Table S1).

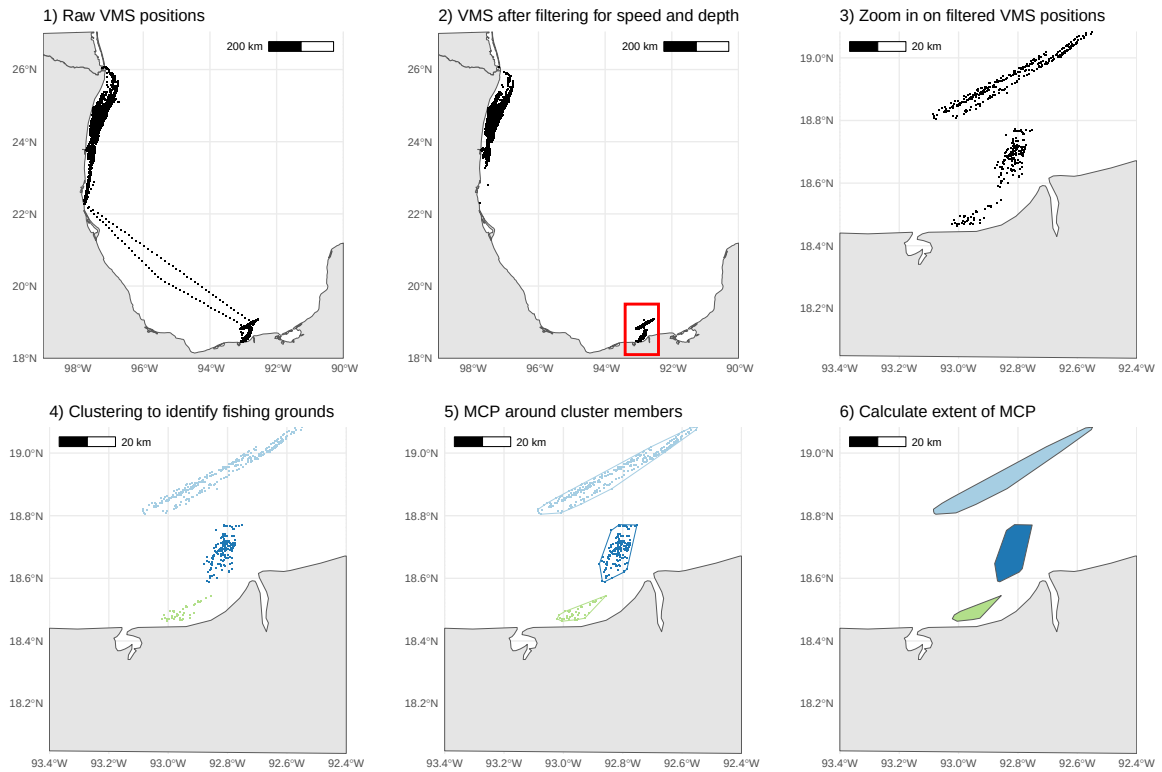


Figure 7: Example showing the process of identifying fishing grounds. Panel 1) shows raw VMS positions for a single vessel. Panel 2) shows VMS positions after filtering for speed and depth parameters. Panel 3) zooms in on the southernmost fishing grounds highlighted in the red square for panel 2). Panel 4) shows the output from clustering with DBSCAN. Panel 5) shows the MCP around cluster members and Panel 6) builds shows the extent of each fishing ground.

5.3 Empirical strategy

5.3.1 Changes in subsidy status

Subsidy allocations are uncorrelated with the outcomes of interest (fishing hours, fishing area, and fisheries production) so we can use these quasi-random changes in subsidy status to test for changes in fishing behavior and fisheries production for economic units whose subsidy status changed at least once in our study period (2011-2019). We estimate the semi-elasticity (*i.e.* the % change in outcome of interest caused by change in subsidy status) of time fishing, fished area, and landings with respect to subsidy status in a two-way fixed-effects regression framework.

$$\mathbb{1}(y_{it} > 0) = \beta S_{it} + \omega_i + \lambda_{r(i)t} + \epsilon_{it} \quad (1)$$

Where $\mathbb{1}(y_{it} > 0) = 1$ if economic unit i has fishing hours, fished area, or landings reported for year t . Then, S_{it} is a dummy variable indicating subsidy status, ω_i and $\lambda_{r(i)t}$ are fixed-effects by economic unit and by region-year. ($r(i)$ denotes the (time-invariant) region to which economic unit i belongs). Our parameter of interest is β , which captures the extensive margin effects. We also estimate model where our outcome variables are simply y_{it} and $\log(y_{it})$, in which case β yields changes along the intensive margin and a semi-elasticity, respectively. Our supplementary materials include alternative specifications and sample definitions (See subsection A.3).

5.3.2 Changes in subsidy amount

Recall that our dataset has three types of economic units: those that were never subsidized, those that were subsidized at least one year, and those that were subsidized every year in our dataset. For the later two types, the *amount* of subsidy they receive varies by year (See Figure 2). This annual variation is due to budgetary constraints, which arise when CONAPESCA receives different amounts of funding in the annual federal budget or when

503 funds are allocated to other programs (*e.g.* aquaculture development). We can use these
 504 quasi-random changes in subsidy amounts to test for changes in fishing behavior and fisheries
 505 production for economic units who were subsidized at least twice between 2011-2019. Like
 506 before, we estimate our coefficient of interest (this time an elasticity) in a two-way fixed
 507 effects framework with our estimating equation taking the form:

$$\log(y_{it}) = \beta \log(s_{it}) + \omega_i + \lambda_{r(i)t} + \epsilon_{it} \quad (2)$$

508 Where $\log(y_{it})$ is the log-transformed outcome of interest and s_{it} is the subsidy amount
 509 allocated to economic unit i in year t , in 2019 Mexican pesos. ω_i and $\lambda_{r(i)t}$ are fixed-effects
 510 by economic unit and by region-year. The parameter of interest is β , which captures the
 511 elasticity of fishing with respect to subsidy amount. Our robustness checks for this analysis
 512 (See subsection A.3) test for changes in the estimated coefficient when limiting the sample to
 513 vessels subsidized at least 3, 4... 8 times (Figure S8) or where we use different specifications
 514 (Figure S9).

515 **5.3.3 An *imromptu* reform**

516 Mexico eliminated fuel subsidies starting in 2020. We first test for changes in the probability
 517 of a vessel exiting the fishery using the following specification:

$$E_{it} = \beta \text{Post}_t + \omega_i + \epsilon_{it} \quad (3)$$

518 Where E_{it} is an indicator variable that takes a value of $E_{it} = 1$ if economic unit i exited
 519 the fishery at year t and is $E_{it} = 0$ otherwise. The variable Post_{it} takes a value of $\text{Post}_t = 1$ if
 520 the observation comes from the post-reform period (2020 and later) and $\text{Post}_{it} = 0$ otherwise.
 521 Finally, ω_i are fixed-effects by economic unit. The parameter of interest is β , which captures
 522 the average annual probability of a vessel exiting the fishery in the post-reform period.
 523 We are also interested in understanding how the probability of exiting the fishery evolved in

the post-reform period. We therefore estimate a model of the form:

$$\mathbb{1}(y_{it} = 0) = \sum_{t \neq 2019} \beta_t \text{Year}_t + \omega_i + \epsilon_{it} \quad (4)$$

Where $\mathbb{1}(y_{it} = 0)$ Indicates whether an economic unit exits the fishery in year t . Year_t are year indicators (omitting 2019 as the reference category). In this case the parameter of interest, β_t , captures the average probability of an economic unit exiting by year t .

We then use the same model specifications, but this time with outcomes $\mathbb{1}(y_{it} = 0)$ and $(y_{it} = 0)$ that measure extensive and intensive margin changes, respectively. This time the sample is restricted to economic units that did not leave the fishery.

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A Supplementary Materials

A.1 Subsidy program description

For the time period analyzed in this study (2011 - 2019), four related fuel subsidy programs in Mexican fisheries have been implemented: *PROCAMPO para vivir mejor* (2011 - 2012), *PROCAMPO Productivo* (2013), *Fomento a la productividad pesquera* (2014 - 2019) and *Subcomponente diesel marino* (2018). The operational rules of the fuel subsidy program in Mexican fisheries are as follows. The fuel subsidy program provides a 2-peso per liter subsidy over a portion of the total fuel used by a vessel, here termed the fuel cap of vessel i ($\hat{Q}_i$). As stated in the program's operational rules⁶, the subsidized portion of fuel for any diesel-consuming vessel is calculated using the following formula:

$$\hat{Q}_i = (MDL_i \times DPC_i) \times AF_i \quad (5)$$

Where $\hat{Q}_i$ represents the fuel cap on the subsidy program given to vessel i . MDL_i denotes the "Maximum Daily Liters" of vessel i , and is what the government expects the vessel's fuel consumption to be. DPC_i represents the "Days Per Cycle": the number of days a vessel is allowed to fish during a fishing season. The MDL_i is based on engine size (Figure S1), while DPC_i is determined by the fishery in which the vessel participates⁷. Finally, AF_i is an exogenous adjustment factor set by CONAPESCA and takes values between 0 and 1. This is independent of fishery, engine power, or stock status and is instead determined by budgetary constraints. The adjustment factor was typically set between 0.4 and 0.7, but local officials may downward adjust it. These variations in adjustment factor provide the source of variation that we will use to identify the effect of fuel fishery subsidies on exacerbating overfishing.

⁶See Section 4.1.2 of Acuerdo por el que se dan a conocer las Reglas de Operación de los Programas de la Secretaría de Agricultura, Ganadería, Desarrollo Rural, Pesca y Alimentación, available at https://dof.gob.mx/nota_detalle.php?codigo=5228713&fecha=30/12/2011#gsc.tab=0

⁷A fishery is defined as the combination of species and location. For example a vessel targeting tuna in the Pacific ocean is part of the Pacific tuna fishery.

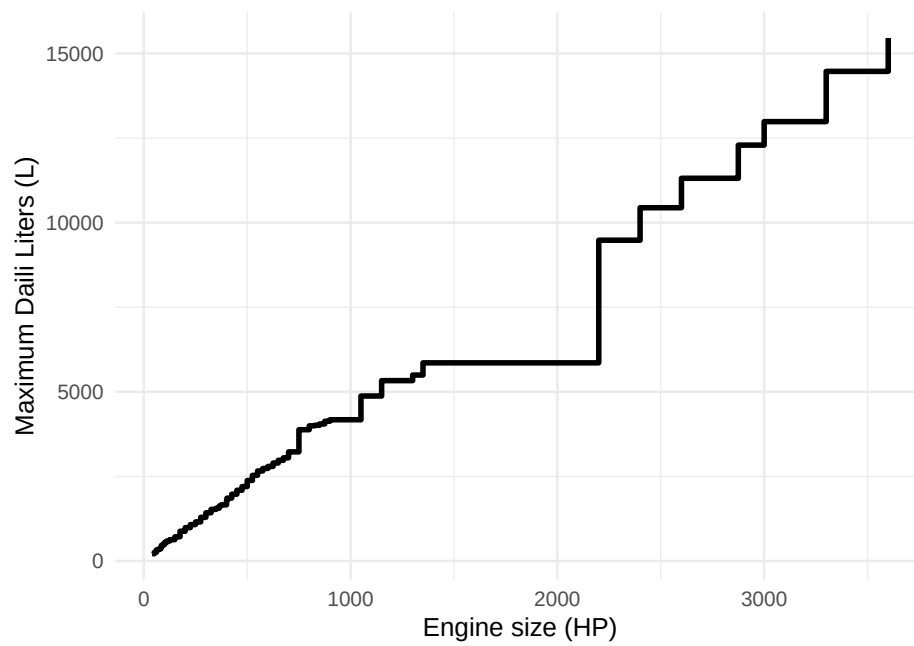


Figure S1: Expected daily fuel consumption for different engine powers. The x-axis shows engine power bins (in HP) as defined by CONAPESCA's operational rules. The y-axis shows the estimated maximum daily liters of fuel to be consumed for the corresponding engine power bin.

A.2 Supplementary Figures and Tables

Table S1: Summary statistics comparing the mean, standard deviation, and range of subsidy amounts and outcome variables across treatment statuses.

	treated	Mean	SD	Min	Max
Subsidy amount (2019 MXN)	Not subsidized	0.00	0.00	0.00	0.00
	Subsidized	790 634.44	1 511 080.57	1521.99	19 633 910.95
Fishing activity (hours)	Not subsidized	1177.34	2405.59	0.00	26 520.33
	Subsidized	4167.75	7053.72	0.00	65 835.78
Fished area (km ²)	Not subsidized	6168.02	9275.14	0.00	92 348.65
	Subsidized	17 966.41	16 374.48	0.00	116 771.44
Landings (Kg)	Not subsidized	9502.86	25 384.33	0.00	285 093.00
	Subsidized	67 204.62	101 070.81	0.00	1 099 556.00

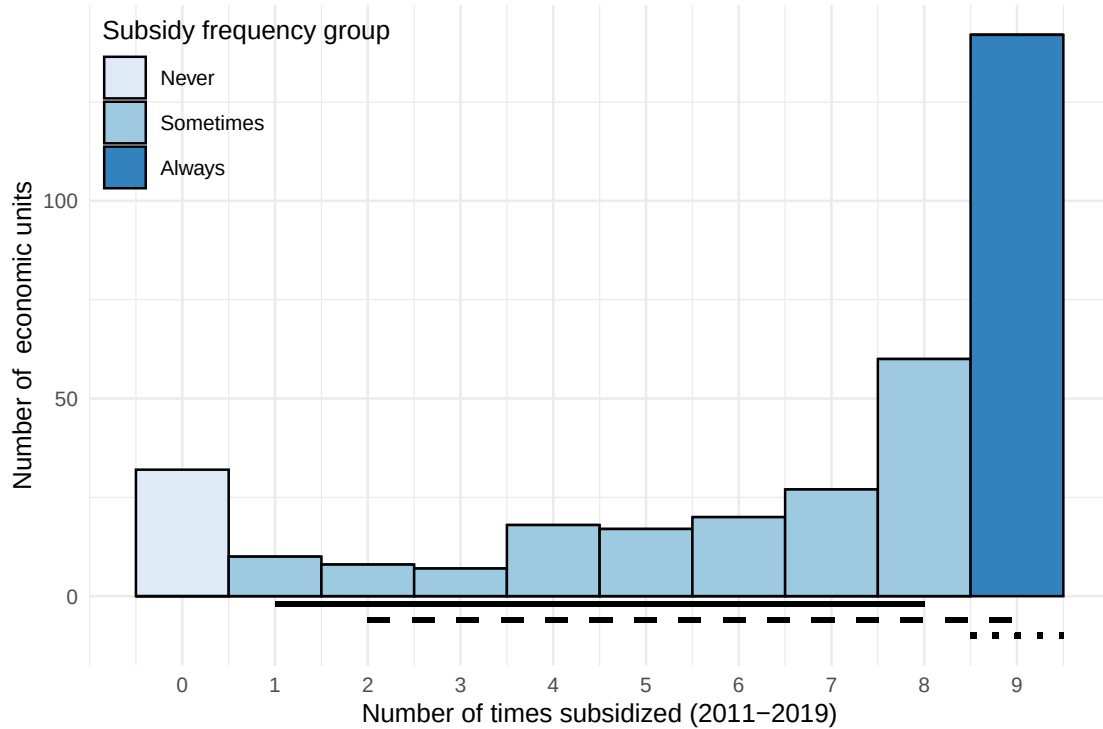


Figure S2: Histogram of frequency with which economic units are subsidized (2011-2019). A value of $N = 0$ along the horizontal axis implies never subsidized, while $N = 9$ implies always subsidized. Our main semi-elasticity estimates use vessels sometimes subsidized (i.e. $N = 1-8$). Our main elasticity estimates use all vessels subsidized $N \geq 2$ times.

A.3 Robustness tests

A.3.1 Responses to changes in subsidy status

We inspect the robustness of our main-text estimates by performing a series of tests using different specifications and samples. Our first test drops all fixed-effects, and instead incorporates covariates for number of vessels in 2011, total engine power in 2011, log-price of diesel fuel, and nino3.4 index interacted by region. This specification produces in similar coefficient estimates. The second test uses the same specification as our main-text estimates, but instead modifies the sample. Recall that our main-text sample excludes economic units never ($N = 32$) and always ($N = 142$) subsidized between 2011 and 2019. This second test includes all economic units, and yields similar coefficient estimates. Our final test removes vessels that were listed in the “fleet modernization” roster, meaning that, in addition to fuel subsidies, they received support to purchase new engines. Removing these vessels yields similar coefficient estimates, leaving our conclusions unchanged. We perform the same battery of tests for our extensive margin estimates (Figure S3 and Table S2), intensive margin estimates (Figure S4 and Table S3), and semi-elasticity estimates (Figure S5 and Table S4).

As an additional check, we test for changes in the estimated coefficients when limiting the sample to vessels subsidized at least 3, 4... 8 times. Figure S2 provides more information on the number of vessels in each sub-sample. We find that the coefficients for time fishing and fished area are largest for vessels subsidized at most 2 times, and gradually converge toward our main-text estimates. The largest coefficient for landings is observed for vessels subsidized just once, but the coefficients also converge toward the main text estimates. Results for extensive margin effects are shown in Figure S6, results for intensive margin effects are shown in Figure S7, and results for semi-elasticity estimates are shown in Figure S8.

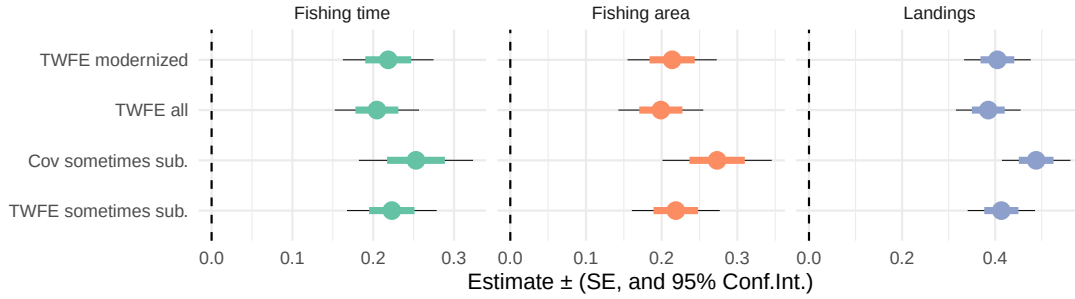


Figure S3: Coefficient estimates for the extensive margin on time fishing, fishing area, and landings with respect to subsidy status for different specifications and samples. Points are coefficient estimates, colored lines show panel-robust standard errors, and black lines show 95% confidence intervals. “TWFE sometimes sub.” refers to the main-text estimates, “Cov sometimes sub” refers to estimates for a model specification that swaps fixed-effects for covariates. “TWFE all” refers to the same two-way fixed effects specification as in the main text, but this time including all economic units regardless of their subsidy frequency and status. “TWFE modernized” removes vessels that were recipients of the fleet modernization subsidies. See Table S2 for more details.

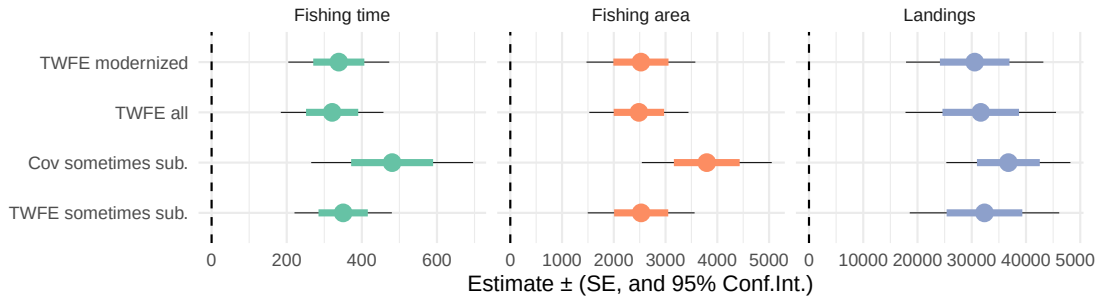


Figure S4: Coefficient estimates for the intensive margin (in levels) of time fishing, fishing area, and landings with respect to subsidy status for different specifications and samples. Points are coefficient estimates, colored lines show panel-robust standard errors, and black lines show 95% confidence intervals. “TWFE sometimes sub.” refers to the main-text estimates, “Cov sometimes sub” refers to estimates for a model specification that swaps fixed-effects for covariates. “TWFE all” refers to the same two-way fixed effects specification as in the main text, but this time including all economic units regardless of their subsidy frequency and status. “TWFE modernized” removes vessels that were recipients of the fleet modernization subsidies. See Table S3 for more details.

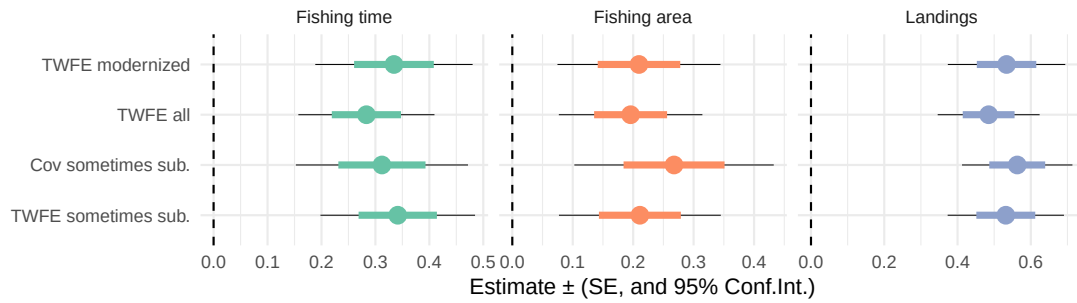


Figure S5: Coefficient estimates for the semi-elasticities of time fishing, fishing area, and landings with respect to subsidy status for different specifications and samples. Points are coefficient estimates, colored lines show panel-robust standard errors, and black lines show 95% confidence intervals. “TWFE sometimes sub.” refers to the main-text estimates, “Cov sometimes sub” refers to estimates for a model specification that swaps fixed-effects for covariates. “TWFE all” refers to the same two-way fixed effects specification as in the main text, but this time including all economic units regardless of their subsidy frequency and status. “TWFE modernized” removes vessels that were recipients of the fleet modernization subsidies. See Table S2 for more details.

Table S2: Effect of receiving a fuel subsidy on time fishing (hours) >0 , fishing area (km^2) >0 , and landings (kg) >0 .

	Fishing time	Fishing area	Landings
A) Main text specification			
Subsidized	0.223 (0.028)***	0.219 (0.029)***	0.413 (0.037)***
N	1431	1431	1431
R^2 Adj	0.531	0.547	0.597
B) Covariates but no fixed effects			
Subsidized	0.253 (0.036)***	0.273 (0.037)***	0.488 (0.037)***
N	1431	1431	1431
R^2 Adj	0.206	0.188	0.373
C) Main text specification with all units			
Subsidized	0.205 (0.027)***	0.199 (0.028)***	0.385 (0.035)***
N	2941	2941	2941
R^2 Adj	0.520	0.588	0.703
D) Main text specification without modernized units			
Subsidized	0.219 (0.028)***	0.214 (0.030)***	0.405 (0.036)***
N	1411	1411	1411
R^2 Adj	0.531	0.547	0.603

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 1A. Panel B) uses the same sample of vessels subsidized at least once, but removes all fixed effects and adds covariates for number of vessels, total engine power, log-price of diesel fuel, and nino3.4 index interacted by region. Panel C) uses the same two-way fixed effects estimation as in A), but includes all vessels in our sample, regardless of number of times subsidized. Panel D) uses the same two-way fixed effects estimation as in A), but removes vessels that received a fleet modernization subsidy.

Table S3: Effect of receiving a fuel subsidy on time fishing (hours), fishing area (km²), and landings (kg).

	Fishing time	Fishing area	Landings
A) Main text specification			
Subsidized	350.248 (65.644)***	2528.599 (523.001)***	32347.423 (6977.661)***
N	1431	1431	1431
R^2 Adj	0.924	0.792	0.710
B) Covariates but no fixed effects			
Subsidized	480.622 (109.120)***	3796.464 (635.744)***	36753.146 (5792.281)***
N	1431	1431	1431
R^2 Adj	0.752	0.488	0.377
C) Main text specification with all units			
Subsidized	320.923 (69.470)***	2485.116 (487.072)***	31658.543 (7053.213)***
N	2941	2941	2941
R^2 Adj	0.956	0.863	0.915
D) Main text specification without modernized units			
Subsidized	338.614 (68.040)***	2524.252 (532.600)***	30543.733 (6410.022)***
N	1411	1411	1411
R^2 Adj	0.922	0.793	0.712

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 1B. Panel B) uses the same sample of vessels subsidized at least once, but removes all fixed effects and adds covariates for number of vessels, total engine power, log-price of diesel fuel, and nino3.4 index interacted by region. Panel C) uses the same two-way fixed effects estimation as in A), but includes all vessels in our sample, regardless of number of times subsidized. Panel D) uses the same two-way fixed effects estimation as in A), but removes vessels that received a fleet modernization subsidy.

Table S4: Effect of receiving a fuel subsidy on time fishing (hours), fishing area (km²), and landings (kg).

	Fishing time	Fishing area	Landings
A) Main text specification			
Subsidized	0.341 (0.073)***	0.211 (0.068)***	0.532 (0.080)***
N	1290	1235	1192
R^2 Adj	0.726	0.700	0.757
B) Covariates but no fixed effects			
Subsidized	0.312 (0.081)***	0.268 (0.084)***	0.563 (0.076)***
N	1292	1240	1196
R^2 Adj	0.305	0.245	0.294
C) Main text specification with all units			
Subsidized	0.283 (0.064)***	0.196 (0.060)***	0.485 (0.071)***
N	2723	2629	2530
R^2 Adj	0.791	0.783	0.846
D) Main text specification without modernized units			
Subsidized	0.334 (0.074)***	0.210 (0.068)***	0.534 (0.081)***
N	1272	1217	1173
R^2 Adj	0.727	0.701	0.756

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 1C. Panel B) uses the same sample of vessels subsidized at least once, but removes all fixed effects and adds covariates for number of vessels, total engine power, log-price of diesel fuel, and nino3.4 index interacted by region. Panel C) uses the same two-way fixed effects estimation as in A), but includes all vessels in our sample, regardless of number of times subsidized. Panel D) uses the same two-way fixed effects estimation as in A), but removes vessels that received a fleet modernization subsidy.

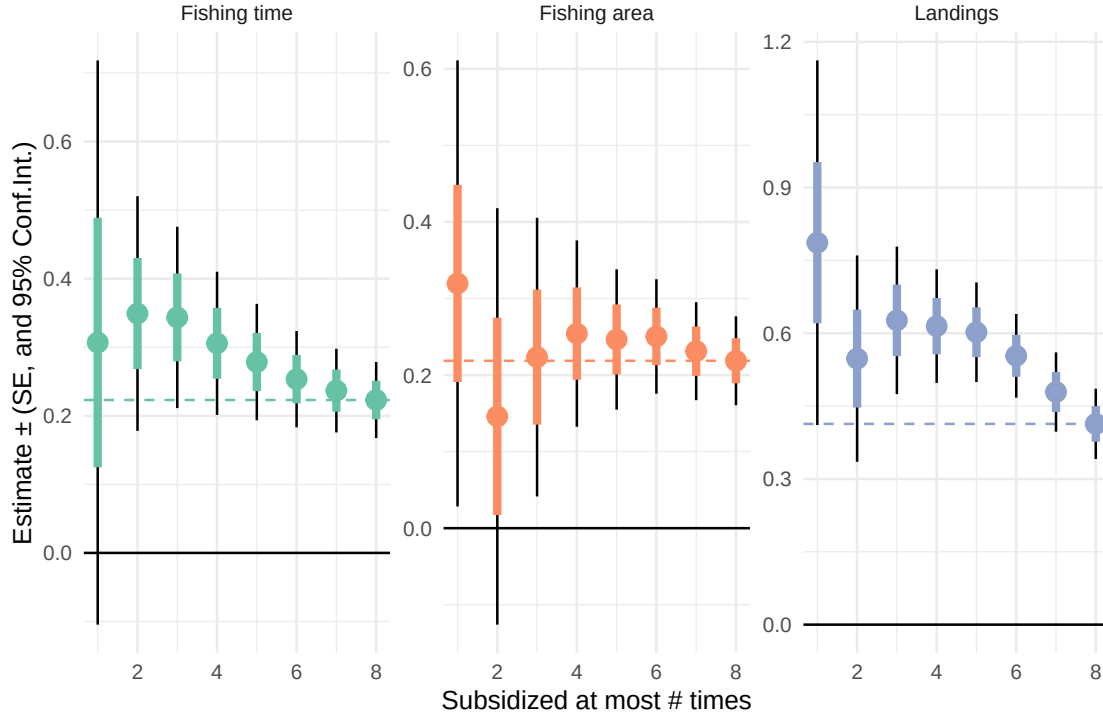


Figure S6: Coefficient estimates for the extensive margin effect on time fishing, fishing area, and landings with respect to subsidy status for different samples based on subsidy frequency. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. Horizontal dashed lines show the coefficient estimate corresponding to our main specification (Table 1). Each point corresponds to a different sub-sample, where economic units are subsidized at most n times, as indicated by the x-axis. In all cases, the rightmost point (subsidized at most 8 times) corresponds to our main-text estimates.

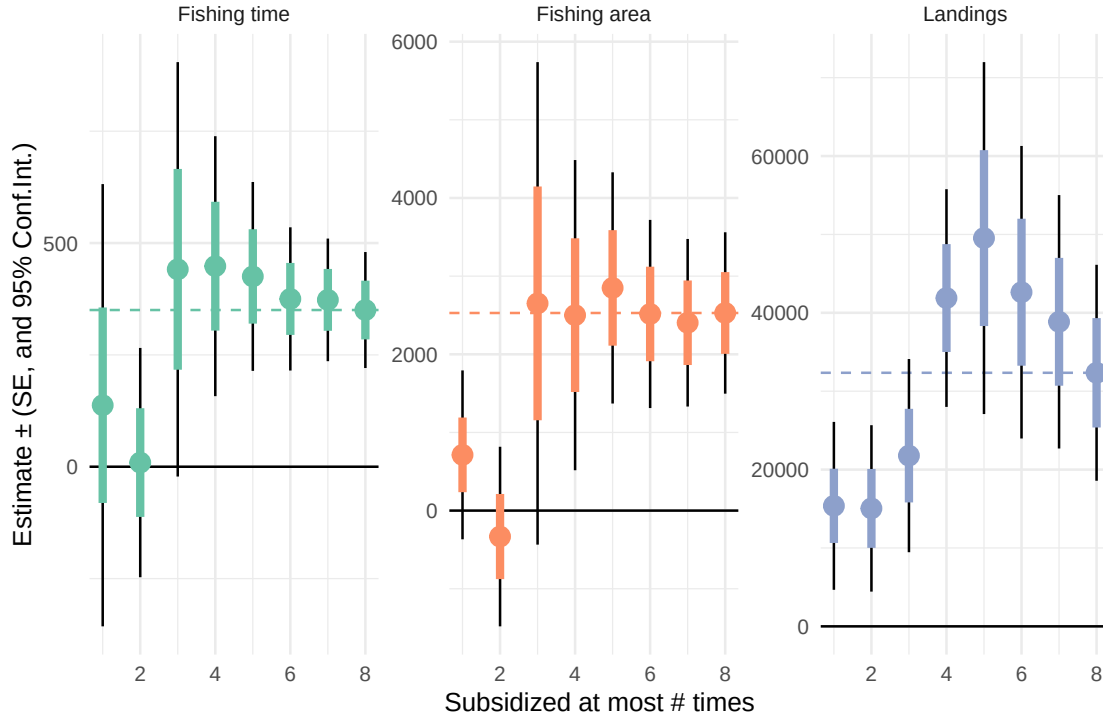


Figure S7: Coefficient estimates for the intensive margin effect (in levels) time fishing, fishing area, and landings with respect to subsidy status for different samples based on subsidy frequency. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. Horizontal dashed lines show the coefficient estimate corresponding to our main specification (Table 1). Each point corresponds to a different sub-sample, where economic units are subsidized at most n times, as indicated by the x-axis. In all cases, the rightmost point (subsidized at most 8 times) corresponds to our main-text estimates.

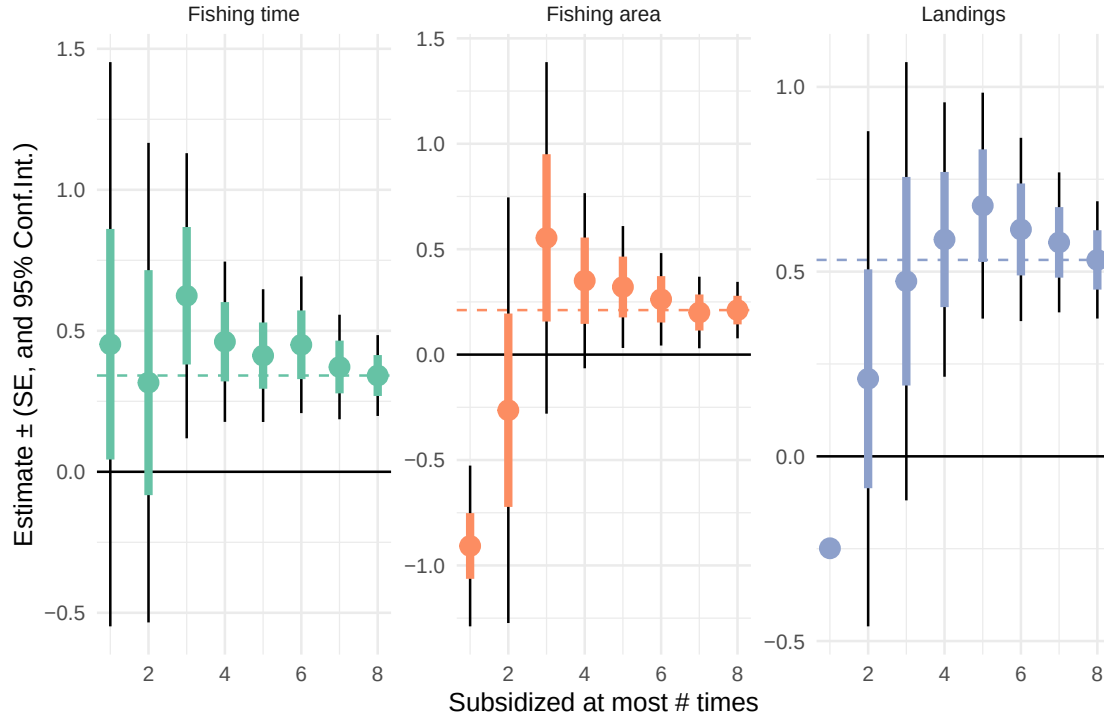


Figure S8: Coefficient estimates for the semi-elasticities of time fishing, fishing area, and landings with respect to subsidy status for different samples based on subsidy frequency. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. Horizontal dashed lines show the coefficient estimate corresponding to our main specification (Table 1). Each point corresponds to a different sub-sample, where economic units are subsidized at most n times, as indicated by the x-axis. In all cases, the rightmost point (subsidized at most 8 times) corresponds to our main-text estimates.

A.3.2 Responses to changes in subsidy amounts

We inspect the robustness of our main-text elasticity estimates by performing a series of tests using different specifications and samples. The data in Table S5 provides model fit statistics and sample sizes, but coefficients are more easily visualized in Figure S9. Our main-text sample includes all economic units subsidized at least twice between 2011 and 2019 ($N = 297$). Our first test modifies the sample to include only economic units that were always subsidized during the same period ($N = 142$). This sub-sample produces similar coefficient estimates, shown in Panel B of Table S5. The next test does the opposite, and excludes economic units that were always subsidized. The results are shown in Panel C of Table S5, and show coefficient estimates similar to those in reported in our main text. As before, the third test removes economic units that participated in the fleet modernization program. This also produces similar coefficient estimates for all outcome variables, as shown in Panel D of Table S5. Finally, we drop our fixed-effects by year and economic unit and instead include the same battery of covariates as before. This specification uses the same sample as our main-text estimates, and results in higher estimates of elasticity of fishing time, fishing area, and landings at 0.2, 0.19, and 0.37, respectively. These estimates are 40-90% larger than our main-text estimates, and are shown in Panel D of Table S5.

As before, we also conduct one more robustness test where we use our main-text specifications but restrict the sample to economic units subsidized at least 1, 2, 3... 9 times. The results are shown in Figure S10. This test yields stable coefficient estimated across all sub-samples.

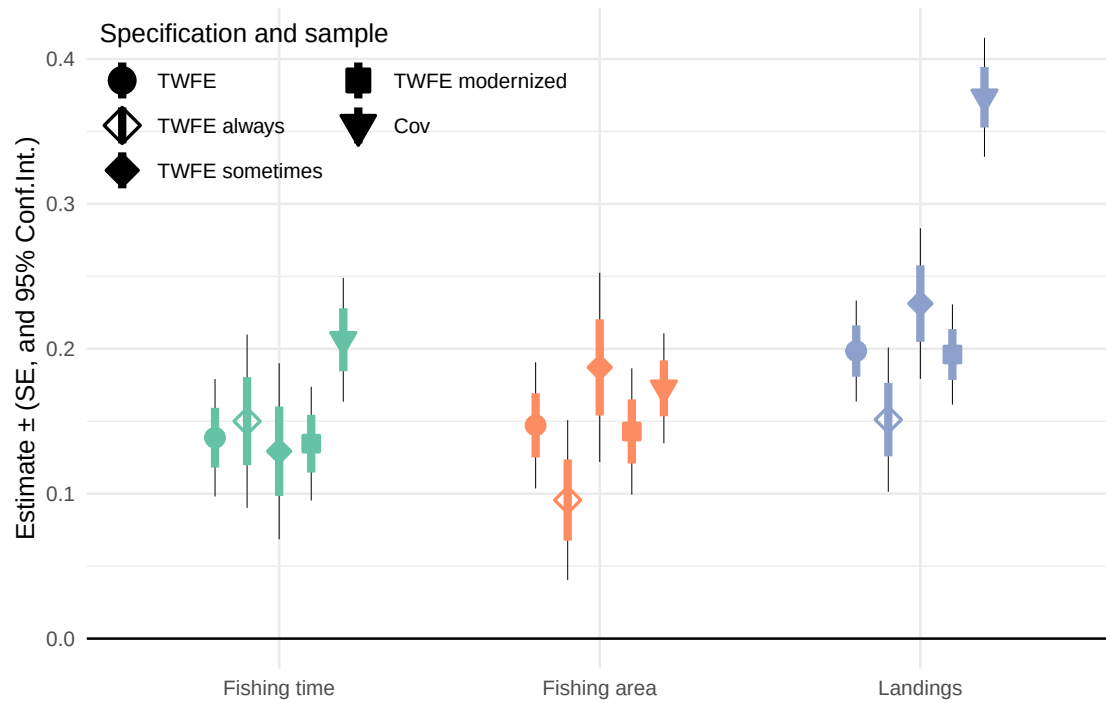


Figure S9: Coefficient estimates for the elasticities of time fishing, fished area, and landings with respect to subsidy amount. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. The main sample combines all economic units subsidized at least twice between 2011 and 2019. Alternative samples, labeled “Modernized” “Sometimes” and “Always”, restrict the sample to economic units that were not part of fleet modernization subsidies, or that are sometimes and always subsidized in the same period, respectively. One-way fixed-effect specifications (labeled “OWFE”) drop year-by-region fixed effects and use annual log-transformed mean national fuel prices, the NINO3.4 index values, and a dummy variable for 2014.

Table S5: Elasticity estimates for time fishing (hours), fishing area (km²), and landings (kg) with respect to changes in subsidy amount.

	Fishing time	Fishing area	Landings
A) Main text specification			
log(subsidy amount[MXP])	0.139 (0.021)***	0.147 (0.022)***	0.198 (0.018)***
N	2240	2202	2246
R^2 Adj	0.850	0.818	0.876
B) Always subsidized			
log(subsidy amount[MXP])	0.150 (0.030)***	0.096 (0.028)***	0.151 (0.025)***
N	1278	1271	1278
R^2 Adj	0.884	0.828	0.905
C) Sometimes subsidized			
log(subsidy amount[MXP])	0.129 (0.031)***	0.187 (0.033)***	0.231 (0.026)***
N	962	931	968
R^2 Adj	0.749	0.746	0.809
D) Removing modernized			
log(subsidy amount[MXP])	0.135 (0.020)***	0.143 (0.022)***	0.196 (0.018)***
N	2208	2170	2214
R^2 Adj	0.850	0.819	0.875
D) Covariates but no fixed effects			
log(subsidy amount[MXP])	0.206 (0.022)***	0.173 (0.019)***	0.374 (0.021)***
N	2242	2206	2247
R^2 Adj	0.517	0.374	0.540

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 2. Panel B) restricts the sample to economic units always subsidized. Panel C) restricts the sample to economic units sometimes subsidized. Panel D) removes economic units that received fleet modernization subsidies. Panel E) uses the same sample of vessels, but removes all fixed effects and adds covariates for number of vessels, total engine power, and nino3.4 index interacted by region.

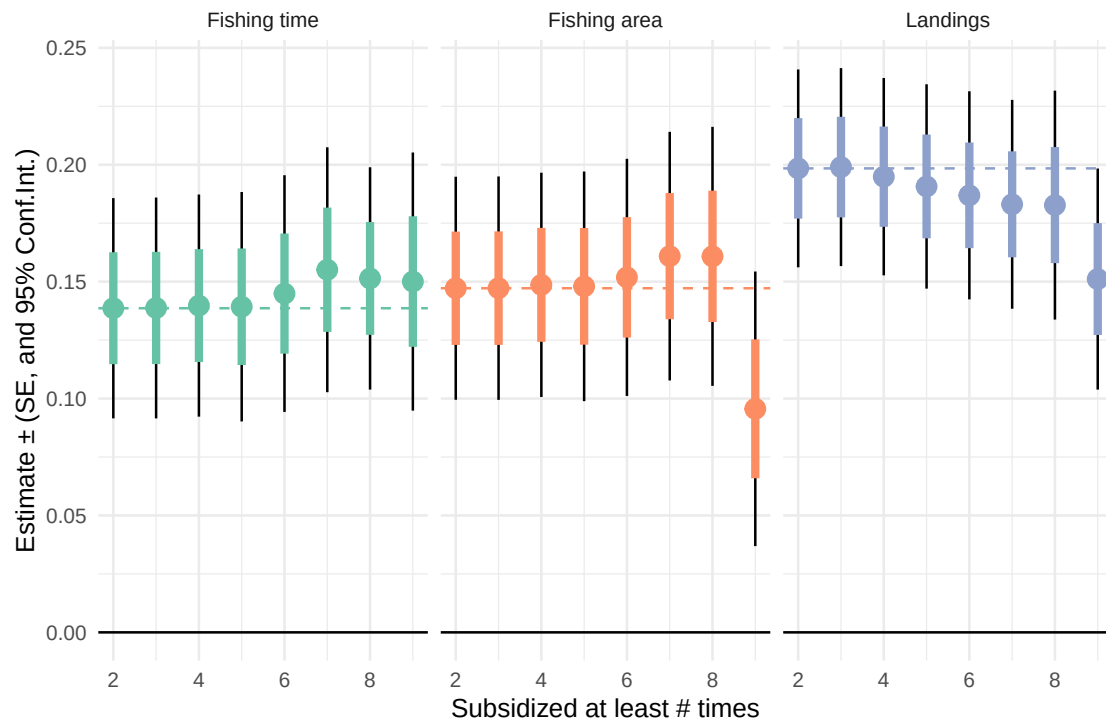


Figure S10: Coefficient estimates for the elasticities of time fishing, fishing area, and landings with respect to subsidy amount for different samples based on subsidy frequency. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. Horizontal dashed lines show the coefficient estimate corresponding to our main specification Table 2. Each point corresponds to a different sub-sample, where economic units are subsidized at least n times, as indicated by the x-axis. In all cases, the leftmost point (subsidized at least twice) corresponds to our main-text estimates.